Missing-functionality module specs — rev 4

MultiCharts .NET · dxFeed (unadjusted @ES) · IBKR — 2026-10-21 (ET)

Changelog

Rev	Date (ET)	Change
1	2026-10-21 12:03	First draft (Analyst 2).
2	2026-10-21	Incorporated Analyst 1 review (`specs/missing-functionality-review-analyst1.md`): added **M-TICK**; listed M-GMM, M-GEX, M-COST, VP helpers as deferred/out of scope; **roll handling** for unadjusted `@ES` (VPF-4 detector, roll-gap removal, AVWAP invalidation, ATR reset, PT+IBKR symbol mapping); **zero new `[Input]`s** — all settings are protected properties (`PriceActionConfig` etc.); `ETraceLevel.Journal` facts with times, not bar numbers; per-FlowQuality-tier delta-z norms; unsupported MC claims turned into probes (function-object laziness, `FullSymbolData.Current → CurrentBar`, bar-end stamping, per-series MaxBarsBack); GV statement corrected (UM p488); PMMS fed by named values; PG p53–54 IOG-off caveat; `EVwapType` extended only with Week/Month/Quarter/RollDate, no `AnchoredEvent`/`AvwapMode`, event manager in `_CustomSignal` via protected `AvwapEvents`; `_PriceActionEngine` rev 2 fixes; 1m/15m acceptance trace; extended Data2 probe.
3	2026-10-21	**M-SWING renamed PRICEACTION** and widened to the full PriceAction scope; all items of `specs/priceaction-improvements.md` (H1–H9, M1–M10, L1–L6) folded into §1 with a review-item → engine → test mapping (§1.7); engine file `_PriceActionEngine.Function.CS` **supersedes `_SwingEngine.Function.CS`** (removed); `SwingConfig → PriceActionConfig` (protected, `enabled=false`); cross-refs (AVWAP lazy dependency, XMKT leader swing, §0 examples) updated. `/workspace/PriceAction` untouched.
4	2026-10-21	User decisions: PRICEACTION defaults stay at PriceAction-library parity; added `EPriceActionPreset {Library, Futures}`, `PriceActionConfig.ApplyPreset()`, `PriceActionConfig.Futures()/Library()` (§1.1); `_VolumeProfileFuturesBase` uses the Futures preset (specified in §1.8, code change deferred until `_CustomSignal` hosts `PriceActionConfig`); Q5 and Q6 resolved; library fixes submitted upstream as PRs #1–#10 (§1.9).

Source module list: `trading-idea-catalogue.md` lines 165–178. Citations: **PG** = MC .NET Programming Guide v1.0 (`mc-docs/01M5BF386FDT0VH0AYQBMXHEF2.pdf`, printed pages), **PT** = Portfolio Trader Manual (`mc-docs/01M5BF39CB6Q6Z75MKY9GA3DF8.pdf`), **UM** = MC User Manual. Base rules: `reviews/analyst-base-class-spec.md` rev 2.7 (D7 minimal inputs; validate once in StartCalc; VPB-9 helper-file convention, `_VpVwapEngine.Function.CS`).

Scope: covered, deferred, out of scope

Item	Status
PRICEACTION, M-AVWAP, M-MTF, M-TOD, M-DAILY, M-XMKT, M-TICK, M-PORT	Specified here
M-NEWS	**Deferred** (user). `NewsBar` anchor reserved and inert; blackout ideas stay gated (F-TM entry window off for those); `_EconomicCalendar` port is a later item
M-GMM (bootstrap, KDE sweep, GMM/BIC, entropy, KL; 9 ideas)	**Deferred / next batch** — those ideas stay gated
M-GEX (external daily file; 1 idea)	**Deferred**; when done, as an ASCII QuoteManager symbol added as Data n (M-XMKT/M-DAILY stream)
M-COST	**Not a module** (rev 2: CS-2 removed). Mapping: MC native Commission/Slippage + the optional cost filter (Rev 2.2 Q1) + G0-a done offline from the journal (dxFeed signal price vs IB fill, Rev 2.7 §A6)
VP-base helpers (range profile [t0,t1], weekly profile stream, per-bar profile ladder, entropy/Parkinson/V(p)/T(p), A2-33 harness)	**Out of scope** of this document — VP-base items

0. Cross-cutting rules

0.1 Lazy evaluation (all engines)

1. **Shape.** Every lazily needed engine is a **plain internal sealed class** (VPB-9 convention) created in `StartCalc()` only when required and otherwise held as `null`. This includes all new engines: `_PriceActionEngine`, `_AvwapManager`, `_TodNormaliser`, `_DailyContext`, `_XmktLeg`, `_TickStream`. Whether an *un-accessed* `FunctionSeries` still runs CalcBar is **not documented** (PG p34 only says create and address the value) → **Probe P1**. Until P1 is answered, existing _ATR/_VWAP function objects are created in `Create()` only for always-on features; nothing lazy depends on a function object.
2. **Enable signal = code, not inputs.** Each engine has a protected configuration property with a disabled default, set by a derived strategy in its ctor or StartCalc (`PriceActionConfig.enabled`, `AvwapEvents`, `TodMetrics`, `DailyConfig`, `XmktLegs`, `TickConfig`). **No new `[Input]` on `_CustomSignal`** (D7: inputs show on all 14 derived strategies and can reset saved workspaces). A strategy that wants to sweep a setting exposes its own `[Input]` or uses the `_TestN` `Variant` index.
3. **Dependencies are lazy and transitive:** enabling M-AVWAP swing/BOS anchors enables (creates) a PRICEACTION instance (`_PriceActionEngine`, candles off unless the strategy enables them); climax anchors enable M-TOD Volume; nothing else.
4. **Lifecycle.** `csharp protected PriceActionConfig PriceActionConfig { get; } = new PriceActionConfig(); // enabled = false protected override void StartCalc() { base.StartCalc(); if (PriceActionConfig.enabled) { m_pa = new _PriceActionEngine { config = PriceActionConfig, trace = TraceLevel >= ETraceLevel.Debug ? s => trace(ETraceLevel.Debug, s) : (Action<string>)null, journal = TraceLevel >= ETraceLevel.Journal ? s => trace(ETraceLevel.Journal, s) : (Action<string>)null }; string err = m_pa.Configure(); // validate once; report, never clamp if (err != null) throw new Exception(err); } else m_pa = null; } protected override void CalcBar() { if (m_pa != null) { if (rollOn(PriceActionConfig.dataStream)) m_pa.OnRoll(BarsOfData(PriceActionConfig.dataStream).Time[0]); m_pa.Step(BarsOfData(PriceActionConfig.dataStream)); } }` Accessors return neutral values (`null`, `NaN`, `Unknown`, empty list) and log once if a rule queries a disabled/unconfigured engine.
5. **Per-bar gating:** commit only on `Bars.Status == EBarState.Close` (PG p20, p35). Dedupe per stream with `bars.CurrentBar` (documented, PG p20). `FullSymbolData.Current` is **not** used (undocumented as an index; FullSymbolData also forces full-series load, PG p17).
6. **StopCalc:** engines → `null`; re-calc starts clean. `requireDataStream()` (existing) checks streams only for enabled engines.

0.2 Logging / journal

`ETraceLevel.Journal` facts (appear with the per-trade journal): swing confirmations, roll resets, AVWAP anchor create/expire/invalidate, MTF first mapping per session, XMKT/TICK stale-leg events, TOD tier resets. Always stamped with **bar times** (`yyyy-MM-dd HH:mm:ss`, chart time zone), never bar indexes (indexes are per stream). Per-bar detail stays at `Debug`; the owner passes a `null` sink when the level is filtered so no strings are formatted. Trade journal lines carry the AVWAP tag `AV:{type}:{anchorTime:yyyyMMddHHmm}` and the swing `confirmTime` used.

0.3 Roll handling (unadjusted dxFeed @ES, rev 2.7)

Continuous data is **unadjusted, always** (Rev 2.2 Q3). `Bars.Info.Name` does not change on @ES, so roll is detected with the **rev 2.7 VPF-4 detector**: session-open gap > k×ATR **and** the date is in the known dxFeed roll-date table. The base exposes `rollOn(ds)` = true on the first bar of the new contract for stream ds. On a roll:

- `_PriceActionEngine.OnRoll()`: clear swings and structure (old price space), reset the TR/ATR ring, skip the splice-bar TR.
- `_DailyContext`: roll-gap removal — difference-adjust prior closes for level series (SMA200, prev H/L/C), ratio for returns (TSMOM, RSI2 change), skip the splice day's TR in ATR20d; `GapPct` uses the adjusted prior close.
- `_AvwapManager`: **invalidate** (not shift) every live anchor of every type; journal `AV invalidated (roll)`.
- `_TodNormaliser`: Range/TR samples of the splice bar are dropped.
- Any Wilder ATR (Daily) restarts its seed.
- **Execution:** Portfolio Trader / chart auto-trading uses IBKR with **Symbol Mapping** of each @ symbol to the IB front contract, re-mapped at each roll (rev 2.7 §E8).

0.4 Recalculation consistency (PG p53–54)

On multi-data charts a signal is calculated on any series' bar close (with IOG, every tick of any series). **With IOG off, orders go out only at Data1's Open tick and "the last calculation before the orders sending at Open is considered valid" (PG p54)** — an entry decided on a Data2-only calc is overridden by the next calc. Rules must therefore be **idempotent per Data1 bar**: derive entry conditions only from closed-bar state that does not change between calcs on the same Data1 bar.

1. PRICEACTION — price-action engine (formerly M-SWING)

Renamed from M-SWING in rev 3 and widened to cover the whole PriceAction library scope (candles + pivots + structure + breakouts). **File:** `/workspace/MultiCharts.Net/_PriceActionEngine.Function.CS` (DRAFT rev 3, uncompiled; must pass CS-0 helper-file compile, C# 7.3). It **supersedes `_PriceActionEngine.Function.CS`**, which was removed (it was never in `PLStudies.csproj`); the Coder must add `<Compile Include="_PriceActionEngine.Function.CS" />` when compiling. The user's library `/workspace/PriceAction` is reference only and is **not modified**.

Origin. `github.com/ettiennehugo/PriceAction` (`PriceActionAnalysis.cs`). The review `specs/priceaction-improvements.md` (items H1–H9, M1–M10, L1–L6) is folded in below; §1.7 maps every item to the engine.

1.1 Types and configuration

- Types: `EPivotMode {Fractal, AtrZigZag}`, `EPivotTieRule {Strict, LeftInclusive}`, `EPivotLabel {None, HH, LH, HL, LL}`, `EStructure`, `[Flags] ECandle`, `PriceActionConfig`, immutable `_Pivot`, internal `_Ring<T>`, `_PaBar`, `internal sealed class _PriceActionEngine`.
- **`PriceActionConfig`** (protected property on `_CustomSignal`, no `[Input]`; sweep via the strategy's own input or `_TestN` Variant): `enabled=false, dataStream=1, pivotMode=Fractal, tieRule=Strict, layers=2, strength=1, atrLength=20, volumeLength=20, volumeIncludesCurrent=true, atrMultiple=1.5, factorDoji=0.1, factorClose=0.1, factorStrong=1.0, factorEpsilon=0.05, equalFactor=0.1, lookbackPivots=5, minBreakoutLayer=1, breakoutNeedsStrong=true, history=50, candles=true`. Defaults equal PriceAction's constants.
- **Presets (rev 4):** `EPriceActionPreset { Library = 0, Futures = 1 }`. `Library` (default) = PriceAction-library parity: `tieRule=Strict`, `volumeIncludesCurrent=true`. `Futures` = `tieRule=LeftInclusive` (flat tops on tick-rounded prices form pivots), `volumeIncludesCurrent=false` (volume norm from prior bars only). Switch with one call: `PriceActionConfig.ApplyPreset(EPriceActionPreset.Futures)` or start from `PriceActionConfig.Futures()`; the preset only touches the fields it owns, so later per-field overrides still apply.
- `Validate()` returns all bad values (ranges as in PriceAction setters, plus `minBreakoutLayer < layers`, `history ≥ lookbackPivots+1`); owner throws in StartCalc. Config is frozen after `Configure()`.

1.2 Lifecycle, laziness, MC integration

- Created in `_CustomSignal.StartCalc()` only if `PriceActionConfig.enabled` **or** a lazy dependent requires it (M-AVWAP swing/breakout anchors, M-XMKT leader swing); otherwise `null` (§0.1). One instance per (stream, config).
- `Step(IInstrument bars)`: ignores ticks unless `bars.Status == EBarState.Close`; dedupes by `bars.CurrentBar`. `Add(barNumber, time, o, h, l, c, v)` is the MC-independent entry point used by tests; out-of-order bar numbers are rejected (warn once); bars are keyed by **bar number, never DateTime** (same-second bars on tick/seconds charts are distinct).
- Roll: owner calls `OnRoll(time)` on the VPF-4 roll bar before `Step` → window, TR/ATR and volume rings and pivots cleared, splice-bar TR skipped; sequence numbers keep increasing (§0.3).
- Sinks: `journal` (ETraceLevel.Journal: pivot confirmations with seq/label/EQ, breakouts, BOS/CHoCH, roll resets; times only), `trace` (Debug: per-bar candle flags), `warnOnce`; owner passes `null` when the level is filtered.

1.3 Indicators (O(1), no per-bar allocation)

- **ATR** = simple average of TR over `atrLength` (PriceAction/`AvgTrueRange` parity), running sum over a ring. `Atr` = NaN until warm; `IsWarm` = both TR and volume rings full.
- **Volume** mean and population σ from running Σv, Σv² over `volumeLength` (parity). `volumeIncludesCurrent=false` computes the norm from prior bars only. Optional `volumeZOverride` (e.g. M-TOD clock-minute volume z) replaces the internal z — fixes the "opening bars always +2SD" bias.
- Storage: fixed `_Ring<T>` buffers sized to the real lookback (bar window = 2·strength+1, TR/volume rings, `history` pivots per layer per side). The only per-bar allocation is one `_Pivot` per confirmed pivot.

1.4 Candle patterns (pure static `ClassifyCandle(bar, prior, atr, config)`)

Evaluated on the closed bar only. Rules (Brooks, *Trading Price Action Trends* p.89, where cited):

Flag	Rule (ATR-scaled: near = factorClose·ATR, eps = factorEpsilon·ATR)	Change vs PriceAction
Bull / Bear	close > / < open	—
Doji	|o−c| ≤ factorDoji·ATR	—
BodyAtHigh (was `highPin`)	range > 0, body/range < `SMALL_BODY_RATIO` (1/3), high − body top < eps → hammer, long lower tail	renamed; named constant; body top = max(o,c) for both colours
BodyAtLow (was `lowPin`)	mirror → shooting star, long upper tail	renamed
Inside / Outside	strict vs prior high/low	—
Engulfing	body engulfs prior **range** (o < prior low and c > prior high, mirror for bear)	documented as range-engulfing
BullReversal	prior bear, this bull, not doji, ≥ 2 of: (1) open ≤ prior close + near and close > prior close; (2) lower tail ∈ [1/3, 1/2]·range and upper tail ≤ near; (5) close > prior high. Rules 3–4 not implemented	rule 2 fixed (PriceAction accepted a zero tail); split bull/bear flags
BearReversal	mirror; rule 1 = open ≥ prior close − near and close < prior close − near ("well below")	rule 1 fixed (PriceAction used prior open)
Strong	body ≥ factorStrong·ATR, opens within near of one extreme, closes within near of the other	—
Vol*	z = (v − mean)/σ (or override): AboveAvg z>0, Plus1/2/3SD, BelowAvg, Minus1/2/3SD	z-based; σ=0 → no volume flags

Flag	Rule (ATR-scaled: near = factorClose·ATR, eps = factorEpsilon·ATR)	Change vs PriceAction
BreakoutUp / Down	see §1.6	fixed

`Candle` = flags of the last closed bar; `Changed` = flags that differ from the previous bar ([Flags] XOR), replacing the noisy `StateChanged()`.

1.5 Pivots, labels, structure

- **Fractal** layer 0: mid bar of a 2·strength+1 window; `tieRule=Strict` (>/<, PriceAction parity — adjacent equal highs never pivot) or `LeftInclusive` (`mid ≥ left && mid > right`, catches flat tops). **AtrZigZag**: starts undecided, tracks both extremes, direction set by the first ≥ atrMultiple·ATR move.
- Layer k: middle of the last three confirmed layer k−1 same-side pivots (same tie rule), confirmed on the bar that confirmed the newest of the three — no lookahead.
- Every pivot is **immutable**: `seq` (per layer and side, assigned at confirmation, monotonic across rolls, **never renumbered** — replaces `renumberPivots`), price, extremeTime/Bar, confirmTime/Bar, label, isEqual, equalToSeq, legSizeAtr.
- **Labels on every layer**: always HH/LH (highs) or HL/LL (lows) vs the previous same-side pivot; ties → LH/LL. No threshold gap (PriceAction required |Δ| > factorClose·ATR, leaving pivots unlabelled).
- **Equal / double top/bottom:** `isEqual` if within equalFactor·ATR of any of the last `lookbackPivots` same-side pivots of that layer; `equalToSeq` names the match. **The earlier pivot is never modified**, and HH/LH is kept.
- Structure per layer: Up (HH+HL), Down (LH+LL), else Range.

1.6 Breakout and BOS

- **Breakout** (candle flag): bar must be Strong (if `breakoutNeedsStrong`) and Bull/Bear; **close** beyond the last pivot of a layer ≥ `minBreakoutLayer` (default 1) whose `seq` has **not been broken yet**; checked from the highest layer down; each pivot breaks once. Fixes PriceAction's "every strong bar is a breakout with ≤ 2 layers" and its skipping of layers 0–1.
- **BOS** (layer 0, any bar): close beyond the last swing high/low not yet broken; **CHoCH** = BOS against the structure as it was before the break. 60-min BOS ideas use an instance on the Data2 stream.

1.7 Review-item mapping and tests

Unit tests run `Add()` with CSV fixtures (no MC needed): `es_1m_normal.csv`, `es_1m_roll_day.csv` (unadjusted splice), `flat_tops.csv`, `gaps.csv`, plus hand-built 5–10-bar cases. Golden files = journal lines.

Review item	How the engine addresses it	Test / acceptance
H1 non-pivots in pivot lists	Pivots appended only when confirmed, on closed bars; never on a flag flip	Tick-replay of a bar that makes then un-makes a high: pivot list unchanged
H2 constant `index` after 500 bars	Bar number supplied by caller (`CurrentBar`); pivot `seq` monotonic	2000-bar run: bar numbers and seq strictly increasing
H3 forming-bar repaint	Commit only on `Status == Close`; no provisional state	Closed-bar replay == IOG tick replay == playback (identical journal)
H4 `renumberPivots` asymmetry/index-0/rewrite	Removed; immutable `seq` at confirmation	Pivot fields never change after creation (snapshot compare over the run)
H5 double-top relabels older pivot	`isEqual+equalToSeq` on the new pivot only; labels kept	Two equal highs: first keeps HH/LH, second has isEqual and equalToSeq = first.seq
H6 breakout bugs	Explicit rule §1.6, unbroken-pivot tracking	layers=2: strong bar without close beyond pivot → no breakout; same pivot cannot break twice
H7 `Clear()` stale state	`Reset()` clears every field	Reset mid-run == fresh engine on remaining bars
H8 `Size` setter leaves stale pivots	No window-size setter; rings sized at Configure, config frozen	n/a (API removed); Configure twice → independent state
H9 out-of-order / duplicate timestamps	Keyed by bar number; out-of-order rejected with warning	Two bars with equal time, different numbers → both processed; decreasing number → ignored + warn
M1 O(n) ATR/volume, allocation	Running sums over rings	ATR/volume equal a naive recompute within 1e-9; allocation profiler: 0 B/bar excluding pivots
M2 SortedList RemoveAt(0)	`_Ring<T>` fixed buffers	Benchmark 1M bars: linear time

Review item	How the engine addresses it	Test / acceptance
M3 no roll handling	`OnRoll()` resets; splice TR skipped	Roll-day fixture: ATR after roll equals ATR of new contract only; no pivot spans the splice; journal "roll reset"
M4 silent setters	`Validate()` reports all; config frozen	Each bad value → error text contains its name
M5 HH/LH threshold gap; layer 0 only	Always labelled, per layer, ties documented	Equal-price pivot → LH and isEqual; layer 1 labels present
M6 strict comparator	`EPivotTieRule` option, Strict default	`flat_tops.csv`: Strict → no pivot, LeftInclusive → pivot at left bar
M7 pin naming, magic constant, reversal rules	`BodyAtHigh/Low`, named constants, corrected rules 1–2	Hand-built bars for each rule (zero-tail bull → not rule 2; bear close between prior close and prior open → not rule 1)
M8 mutable public API	Read-only properties, immutable `_Pivot`, no collections exposed	Compile-time (no setters)
M9 no tests / order dependency	Pure `ClassifyCandle`; `Add()` entry; fixtures above	Test project (Coder)
M10 DLL deployment, no MC awareness	Source helper file in PLStudies, `Step(IInstrument)` adapter	CS-0 compile; MC acceptance trace below
L1 stale comment	n/a in new code	—
L2 generic exception	Owner throws with the Validate text	—
L3 population σ, no TOD	Parity kept, documented; `volumeZOverride` hook	z override wired to M-TOD in one strategy
L4 noisy StateChanged	`Changed` = flag XOR	—
L5 ATR 0 before warm-up	`Atr` NaN, `IsWarm`	First atrLength bars: NaN
L6 duplicated high/low code	Shared `beats()` / side-parameterised paths	Code review

MC acceptance trace (VPB-8): Data1 ES 1-min, Data2 ES 15-min, both Globex 24h; instances on ds1 and ds2, Journal on. Check: (a) each ds2 pivot `confirmTime` is a 15-min close and ≤ the Data1 bar time where it first appears (HTF pivot visible at 10:07 comes from the ≤ 10:00 15-min close); (b) no pivot re-emitted or changed; (c) IOG on/off, backtest vs playback give identical journals; (d) roll date shows "roll reset" and no pivot straddles it; (e) layer-0 Strict pivots on ds1 match PriceAction's layer-0 pivots on the same closed bars, except the documented differences (no renumbering, no relabel, corrected candle rules).

1.8 Futures defaults in `_VolumeProfileFuturesBase` (rev 4)

`_VolumeProfileFuturesBase` (`/workspace/MultiCharts.Net/_VolumeProfileFuturesBase.Strategy.CS`, user code; a different copy exists in `/workspace/mc-strategies`) must use the Futures preset. **Specified, not edited:** `PriceActionConfig` does not exist on `_CustomSignal` yet, so editing the base now would not compile. The minimal change, to be applied together with the `_CustomSignal` integration:

```
// _VolumeProfileFuturesBase.Create() (or ctor), after base.Create():
PriceActionConfig.ApplyPreset(EPriceActionPreset.Futures);   // enabled stays false: still lazy until a derived strategy enables it
```

Derived futures strategies only set `PriceActionConfig.enabled = true` (plus any per-field overrides). Non-futures strategies keep library parity.

1.9 Upstream PriceAction library (rev 4)

The user's library is still not modified by this spec. With the user's authorisation, the review items were submitted as separate PRs against `ettiennehugo/PriceAction` master. They are not merged; no direct pushes to master.

PR	Items	Title
#1	M9	Add xUnit test project with fixtures
#2	M7	Correct reversal-bar rules and name pin/reversal constants
#3	H1, H2	Fix non-pivots left in pivot lists and frozen bar index
#4	H4 a/b	Fix renumberPivots high/low asymmetry and skipped index 0
#5	H6	Fix breakout detection
#6	H7, H8, H9	Fix Clear() stale state, Size setter stale pivots, out-of-order bars

PR	Items	Title
#7	M1, L5	Incremental ATR/volume statistics; IsWarm
#8	M4, M6, L2	Setter validation; optional PivotTieRule (default Strict)
#9	M3	NotifyRoll for unadjusted continuous futures
#10	H3, H5	[BREAKING 2.0] Closed-bar commit model; no double-top relabel

Not submitted: H4c (fixed sequence numbers instead of renumbering), M2 (ring-buffer storage), M8 (read-only collections; breaking), M10 (MC packaging), L1/L3/L4/L6. The MC engine `_PriceActionEngine` already covers these.

2. M-AVWAP — event-anchored VWAP vs F-VWAP

Relation. Same math, one accumulator type (`_VpVwapEngine`: typical price (H+L+C)/3, ΣPV, ΣP²V, ΣV, σ = √(ΣP²V/ΣV − VWAP²), provisional/committed split for IOG).

- **F-VWAP** = up to **3 user-configured slots** on `_VolumeProfileBase` (`Vwap1..3Type`, existing inputs), static or calendar anchors.
- **M-AVWAP** = a **code-configured set of event anchors** with lifecycle (age, touch, expiry, tags), `_AvwapManager`, homed in **`_CustomSignal`** (it needs no VP data, so non-VP strategies can use it).
- `_VWAP.Function.CS` stays as-is for indicators.

`EVwapType` answer. Extend it only for single-anchor calendar resets; **values 0–3 unchanged**, no new input: `EVwapType { Disabled=0, Session=1, Length=2, AdaptiveFromDate=3, Week=4, Month=5, Quarter=6, RollDate=7 }`. Week/Month/Quarter reset at the first bar whose exchange-session date is in a new week/month/quarter; RollDate resets on the VPF-4 roll bar. Implemented as a reset predicate in `_VpVwapEngine`. `Day` is omitted (identical to `Session` on the 24h Globex template). **No `AnchoredEvent` value and no `AvwapMode` input**: a slot holds one value (`CurrentVwapN`), N dynamic anchors do not fit, and event anchors are rule-driven.

Enabling: protected `AvwapEvents` (`[Flags] EAvwapAnchor`, default `None`) + protected `AvwapLimits { maxAnchors=6, maxAgeSessions=10, replaceSameType=true, bandSigmas={1,2,3}, slopeBars=5, dataStream=1 }`. None → manager is null.

Anchor events:

Event	Definition	Dependency
`SwingHigh` / `SwingLow`	PRICEACTION confirmed swing; anchor at the extreme bar; ΣPV from extreme to confirmation back-filled from a per-bar PV ring kept by the manager (ring length = swing strength + 1); usable only from `confirmTime`	lazily enables PRICEACTION (ds of the manager)
`Breakout`	PRICEACTION layer-0 BOS bar (`BosUp/BosDown`) or a `BreakoutUp/Down` candle flag (§1.6)	PRICEACTION
`GapOpen`	first RTH bar with |open − prior RTH close| ≥ 0.5 × `_DailyContext.Atr20d`; fallback if Daily not valid: ≥ 3 × Data1 SMA-TR(20)	M-DAILY if enabled
`VolumeClimax`	bar volume z ≥ 3 by M-TOD Volume (clock-minute norm); fallback while M-TOD not valid: 20-bar volume z ≥ 3	lazily enables M-TOD Volume
`WeekOpen` / `MonthOpen` / `QuarterOpen`	first bar of the period	—
`NewsBar`	reserved, inert (M-NEWS deferred)	—

`RollDate` as an *event* is not needed: a roll invalidates all anchors (§0.3) and calendar RollDate lives in F-VWAP slots.

Per anchor: id, type, anchorTime, ΣPV/ΣP²V/ΣV, value, σ, bands ±k σ, slope (ticks/bar and ATR/bar over `slopeBars` closed bars), age (bars, sessions), `touched` + touch count, `expired`, tag. **Lifecycle:** evict oldest of the same type at `maxAnchors`; expire after `maxAgeSessions`; replace same type when `replaceSameType`; **invalidate all on roll**. Create/expire/invalidate are Journal facts.

3. M-MTF — higher timeframes from MultiCharts multi-data

HTF bars come from extra chart series read through `BarsOfData(n)` (PG p19–20), e.g. Data1 = ES 1-min, Data2 = ES 60-min. No internal bar building (supersedes catalogue lines 1119, 2527 — catalogue owner to update).

1. `HtfClosed(ds, ago)`: if `BarsOfData(ds).Status == EBarState.Close` use index `ago`, else `ago+1`. Default for all HTF rules (`closedBarsOnly`, VPB-8). Forming-bar access only via explicit `HtfForming(ds)`, journaled.
2. HTF engines step only on that stream's closed bars (CurrentBar dedupe) — built into each engine.
3. Mapping: for a Data1 closed bar at t, HTF context = latest closed HTF bar with Time ≤ t. Whether MC stamps bars with their **end** time (so the 10:00 60-min bar is closed and usable on the 10:00 1-min bar) is **not in the PG → Probe P2**. First mapping per session is a Journal fact.
4. Recalc consistency per §0.4.

5. Lookback: whether MaxBarsBack applies per series is **not in the PG → unverified (P2)**; for long HTF/daily lookbacks prefer reading history via `FullSymbolData` (PG p22) at StartCalc/once per HTF bar.
6. Session template: reuse the **VPF-3 StartCalc assertion** (24h CME Globex template, ETH bars present) for every stream; additionally assert same `Info.Name` root for HTF streams.

Probe P2 (VPB-8 acceptance, extended): Data1 1-min + Data2 15/60-min, IOG on/off, backtest with/without Bar Magnifier, playback, live. Log per CalcBar: which stream triggered (Status per stream), `BarsOfData(2).Time[0]`, `Status`, `Close[0]`, `CurrentBar`. Answers: (a) forming Data2 bar values/Status intrabar; (b) whether bar times are **end** stamps; (c) **ordering** of Data1 and Data2 closes sharing a timestamp; (d) the PG p54 "last calc before Data1 Open is valid" rule in practice; (e) per-series MaxBarsBack. Expected in backtest without IOG: closed HTF bar. Code uses `HtfClosed()` regardless.

4. M-TOD — time-of-day normalisers

Recommendation: separate small lazy engine `_TodNormaliser`. Reasons: shared non-trivial state (session-minute × 20-session rings per metric) used by many ideas and by M-AVWAP; must be keyed to the exchange session clock (`_CmeSessionClock`) with holiday/half-day handling (`_CmeHolidays`) — one tested implementation; memory is small but non-zero, so lazy; metric-agnostic `(sessionMinute, value)` input.

Config (protected): `TodMetrics` ([Flags] Volume, Delta, Range, TrueRange; default None), `TodSessions=20`, `TodMinSessions=10`, bucket = clock minute (or bar-size bucket if Data1 > 1 min). **Outputs:** `Z(metric)`, `Median(metric)`, `Pct(metric)`, `ZCum(metric)` (session-to-date vs same-minute cumulative norm); norms from prior **completed** sessions only; half-days excluded; commit on Close. **Delta z per FlowQuality tier:** delta comes from the rev 2.7 chain (MC-BidAsk > Quote > TickRule; BarUpDown never used for signals). Each sample is tagged with its tier; the norm **resets at a tier change** (journaled) and `valid=false` until `TodMinSessions` sessions of the same tier exist. The VPF-5 delta API is still to be implemented.

5. M-DAILY — daily/weekly context (approved)

Source: daily (optional weekly) series on the chart, same unadjusted `@` symbol, read via `HtfClosed`; history read via `FullSymbolData` where long lookbacks are needed. Fallback without a daily stream: aggregate Data1 at `LastBarInSession`. **Config (protected):** `DailyConfig { dataStream=0 (=fallback), weeklyStream=0, atrLength=20, smaLength=200, rsiLength=2, tsmomLookbacks={252,126,63} }`; disabled when no consumer sets `NeedsDaily`. **Outputs:** `Atr20d` (Wilder — documented; seed restarts at roll), `Sma200`, `Rsi2`, `Tsmom(n)`, `PrevHigh/Low/Close`, `PrevRange`, `GapPct`, `PrevWeekHigh/Low`; `valid` false until enough bars. **Roll-gap removal** per §0.3 (unadjusted data). **Settlement vs last trade** on dxFeed daily bars: **unverified** → add to rev 2.7 §E live-check list; until then compare dxFeed daily close vs CME settlement for 5 sessions.

6. M-XMKT — cross-market bar streams

Extra `BarsOfData(n)` series on the chart (NQ for ES, ZN, VIX …), **bar resolutions** (seconds/minutes). **Tick-level legs use M-TICK, not a chart stream:** per PG p53–54 a tick Data n makes CalcBar fire on every tick of that stream, breaking "zero cost unless used". No custom feeds; Global Variables not used — UM p488: GV pair *backtesting* works only in Portfolio Backtester (GV can pass values between charts in real time), so they are unsuitable for reproducible backtests.

1. Stream map as protected `XmktLegs` (role → ds, `maxStaleSec` default 2× bar size); StartCalc logs `Info.Name`, `Resolution`, `DataFeed` and fails loudly if a needed stream is missing.
2. Alignment: last closed bar of stream n with Time ≤ Data1 time (`HtfClosed`); stale → leg unavailable (Journal fact), no trade on its signal.
3. Each instrument keeps its own session template; measures computed on its own bars, never interpolated.
4. Lazy helper per leg: k-bar log return, rolling correlation/beta, RS-spread z, leader swing (own `_PriceActionEngine`).
5. Orders Data1 only; two-leg trading needs M-PORT.
6. Availability of VIX/ZN etc. depends on the dxFeed subscription bundles (rev 2.7 §E1).

7. M-TICK — lazy DataLoader 1-tick engine (new)

Need: catalogue ideas at lines 1280, 1291, 2226, 3341–3347 — seconds bursts, inter-print times, trade-size stream. **Mechanism:** `_TickStream` plain class, created only when a derived strategy sets protected `TickConfig.enabled`. It issues a **DataLoader** 1-tick Trade request for the Data1 symbol (PG p71–75; `Subscribe2RT` for real time), bounded to the dxFeed tick history window (~2 years). No tick chart stream (see §6). **Outputs** (per Data1 closed bar, from ticks with time in the bar): print count, max prints per second, inter-print time median/p10, burst flag (prints/s z vs M-TOD-style norm), trade-size histogram, large-print count ≥ threshold. Each value carries a FlowQuality tag; missing/late tick data → `valid=false` + Journal fact. **Gate:** the **per-trade-size ideas stay parked until rev 2.7 §E6** (dxFeed trade-size fidelity through MC DataLoader) is verified; count/timing outputs may be used earlier. **Recalc/IOG:** tick aggregation commits when the Data1 bar closes; DataLoader callbacks only buffer (thread-safe queue), CalcBar consumes.

8. M-PORT — portfolio sizing/throttles via Portfolio Trader

Cited mechanisms: portfolio calc of all strategies, one bar at a time, chronological; fills on next Data1 open (PG p57–58, §4.6.10). Money management: Initial Capital, Exposure %, Max % Capital at Risk per Position, Margin per Contract %, Potential Loss per Contract; `Portfolio.SetMaxPotentialLossPerContract(...)`; `IPortfolioPerformance` incl. `CalcMaxPotentialLossForEntry(side, contracts, price)` (PG p59); buying-power example and contract cutting (PG p60–62). PMM signal runs after all strategies each step, on the first instrument, and reaches other strategies only through `pmms_*` functions (PT p9–11, p17). `PortfolioEntriesPriority` is **not used** (PT p10: legacy, conflicts with PMM).

Design:

1. Each strategy publishes, per bar, **named values** (`pmm_set_my_named_num`, PT p17): `ret1d` (daily return of its Data1, published at session close), `riskOpen` (open risk $), `riskNew` (risk of a pending entry = stop distance × BigPointValue × contracts), `family` (numeric id), `ideaId`; plus `Portfolio.SetMaxPotentialLossPerContract(stopDistance × BigPointValue)`.
2. `_PortfolioRiskManager` PMM signal (`_PortfolioRiskManager.Strategy.CS`, idle if only one strategy) reads them with `pmms_get_strategy_named_num` and enforces: max open risk % of equity; per-instrument and per-family caps; **ρ ≥ 0.9 book rule** — ρ computed from the published `ret1d` series (rolling 60 sessions), strategies above the threshold share one risk budget (ES/MES/NQ as one); daily book loss stop → deny entries for the session; max concurrent positions.
3. Live: Portfolio Trader auto-trading via IBKR with per-roll Symbol Mapping (§0.3). Single charts: M-PORT inactive; base sizing applies.
4. Constraints: no Bar Magnifier in Portfolio Trader (BM p6) → intrabar-sensitive ideas need fine Data1; portfolio start = first instrument start (PT p10).
5. The .NET PMM base class / `pmms` equivalents are **not in the 2013 PG** → Probe P5 before any M-PORT code (check MC .NET editor PowerLanguage .NET help).

9. Probes and open questions

#	Question	Status
P1	Does an un-accessed `FunctionSeries` run CalcBar? (counter probe)	Open; design does not depend on it
P2	Data2 forming-bar Status/values, bar-end stamping, same-timestamp ordering, PG p54 rule, per-series MaxBarsBack (§3)	Open; VPB-8 acceptance
P3	dxFeed daily bar = settlement or last trade	Open; rev 2.7 §E list
P4	rev 2.7 §E6: trade-size fidelity via DataLoader (gates M-TICK size ideas)	Open
P5	.NET PMM signal API and named-value functions	Open; before M-PORT code
Q1	VPF-5 delta API for M-TOD delta z	To implement (rev 2.7)
Q2	VPF-4 roll-date table source/maintenance and k for the gap test	Per rev 2.7; confirm values
Q3	`_PriceActionEngine` must pass CS-0 compile; run the §1 acceptance trace	Coder
Q4	Catalogue text for M-MTF ("build HTF internally") to be updated by its owner	Catalogue owner
Q5	PRICEACTION defaults	**Resolved (rev 4):** library parity by default; Futures preset used by `_VolumeProfileFuturesBase` (§1.1, §1.8)
Q6	Port the fixes back to the PriceAction library	**Resolved (rev 4):** submitted as upstream PRs #1–#10 (§1.9), not merged
Q8	Merge order of upstream PRs #1–#10 (independent branches off master, small textual conflicts in `PriceActionAnalysis.cs`); whether to take the 2.0 PR #10	User
Q7	Add `<Compile Include="_PriceActionEngine.Function.CS" />` to `PLStudies.csproj` when the draft is compiled (not done, to keep the build green)	Coder

10. Review status

Reviewed by Analyst 1: see `specs/missing-functionality-review-analyst1.md` (2026-10-21). All points in its §1–§6 are adopted in rev 2 (see changelog).